

RESEARCH ARTICLE

A Stochastic Modeling Approach for Interplanetary Supply Chain Planning

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The unique and complex challenges of sustainable interplanetary travel necessitate a novel space logistics paradigm, which is hereafter referred to as the interplanetary supply chain. To lay the groundwork for this anticipated paradigm shift, this work aims to introduce principles and methodologies of supply chain (SC) planning to the field of space logistics. To this end, a stochastic multistage mixed integer linear programming model is proposed that explores the vision of a permanent human presence on Mars from a logistics standpoint. The model optimizes material flows, propellant production, and in-space infrastructure allocations to identify the optimal SC network design. Notably, demand uncertainty at the Mars base is considered, which makes it the first model to investigate the effects of stochasticity in space logistics planning. Several case studies are investigated, which incorporate data on Earth–Mars transfer windows and flight trajectories obtained through the National Aeronautics and Space Administration's trajectory browser. The results reveal numerous interdependencies and, thus, highlight the need for holistic SC modeling approaches that go beyond the isolated mission planning of the extant literature. For instance, the findings show frequent interactions among spacecraft as they regularly engage in division of labor regarding cargo and propellant transportation. Moreover, complex propellant supply networks that involve strategic in-space infrastructure allocation and propellant production on celestial bodies are established. Overall, the developed model provides valuable insights into the planning of a sustainable interplanetary travel, offers a new SC perspective, and lays the foundation for future research in this area.

Introduction

In recent years, interest groups in business and politics have joined the scientific community in their desire to extend human space travel into deep space, specifically to Mars [1–3]. This historically rare alignment of interests is a potent driver of advancements in the space sector and has made the vision of a permanent human presence on Mars more tangible than ever. It is therefore crucial to lay the theoretical foundation and explore new logistics paradigms that enable sustainable interplanetary travel. In the early days of space exploration, missions were planned as self-contained projects and followed the so-called carry-along approach. It requires all resources needed during the mission to be brought along in the spacecraft. The logistical paradigm changed when a permanent presence in space was established. The International Space Station, for instance, requires regular supply runs from Earth (resupply approach), which is a strategy only viable in near-Earth space [4]. Sustaining a permanent base on another planet, however, poses completely new challenges and thus necessitates a novel space logistics paradigm [4–6], which is hereafter referred to as interplanetary supply chain (ISC).

The subject of sustainable interplanetary travel has been of increasing interest to the scientific community. The Massachusetts Institute of Technology (MIT), one of the leading institutions

in this field, has partnered with the National Aeronautics and Space Administration (NASA) in the Space Logistics Project to study issues related to future space exploration efforts. They propose the concept of an ISC, which refers to an integrated logistics architecture that enables continuous and sustainable access to other celestial bodies in the solar system. In this context, they emphasize the need for demand–supply modeling approaches that study the impact of uncertainties [7]. Similarly, other researchers highlighted the limitations of current space logistics paradigms and proclaimed the necessity to investigate ISC approaches [4–6].

However, the field of optimization-based space logistics is still very much in its infancy. The extant literature focuses almost exclusively on the planning of individual missions or mission campaigns in distinct scenarios. Early works feature static models that are concerned with sending small crews on exploratory missions to Mars (e.g., [8–10]). Ishimatsu [11], then, introduced the concept of generalized multi-commodity network flows (GMCNF), which differentiated between the transported materials (e.g., propellant, payload, the spacecraft itself, etc.). Given the clear limitations of static models, a number of time-expanded GMCNF models have been developed (e.g., [4,12,13]). These were also the first models that accounted for in situ resource utilization (ISRU) capabilities and their impact on human Mars exploration. In more recent works, Chen et al.

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[14] specifically study ISRU deployment strategies, and Chen and Ho [15] incorporate in-space infrastructure such as propellant depots in their model.

Notably, interplanetary logistics has not been approached from a supply chain (SC) perspective to date. The extant literature focuses on exploratory missions to Mars in deterministic settings, which reveals substantial research opportunities. Complex ISC network structures, which enable regular deep space transportation to consistently meet demands at a martian base, have not been addressed. This includes important SC considerations such as interactions among spacecraft, the integrated coordination of vehicle fleets, and complex propellant supply networks. Most importantly, the issue of uncertainty has been completely omitted thus far. However, uncertainty can be expected to play a major role in ISC planning given the extremely long transportation times and technical complexity of this endeavor. This is why, according to MIT's Space Logistics Project, demand-supply modeling with uncertainty is considered an integral step toward an "Interplanetary Supply Chain Management and Logistics Architecture" [7].

To address these research gaps, a holistic SC model is proposed that explores the concept of the ISC as a new space logistics paradigm. To this end, a multistage stochastic programming (MSP) model for ISC planning is developed that accounts for demand uncertainty at a Mars base. This modeling approach has proven valuable in many terrestrial SC applications as it is able to incorporate network design and logistics decisions under uncertainty. These characteristics make MSP a good fit for strategic ISC planning, which has similar requirements. The developed model incorporates material flows, propellant production, and inventories while also determining the optimal locations of infrastructure elements like on-orbit depots and ISRU facilities. Notably, the transportation of materials from Earth to Mars and the production/supply of propellant in space is handled in 2 separate but interacting networks, which is referred to as a layered network in the following. Several case studies are investigated, which feature actual data on interplanetary transfer windows and flight trajectories obtained through NASA's trajectory browser. By merging all these principles of interplanetary space flight with a proven SC optimization methodology, this work sets out to establish the concept of ISC planning in the realm of space logistics.

The contributions to the field of space logistics are manifold. The work at hand provides insights into the planning of deep space operations by means of a holistic ISC model that surpasses the limitations of the isolated mission or mission campaign planning in the existing literature. To the best of the author's knowledge, it is the first integrated model that jointly optimizes material transports to Mars and a supporting propellant supply system. As a result, a new perspective on fuel management is offered by incorporating mass-dependent propellant consumption, limited fuel availability, and the optimal allocation of in-space infrastructure to establish propellant supply lines. Moreover, the integration of stochastic demands makes it the first space logistics model to account for uncertainty. Last, the proposed model accounts for orbital dynamics by considering the concept of a dynamic network in which nodes are in perpetual motion. Overall, this comprehensive modeling approach advances the field of space logistics by an SC perspective.

The remainder of the paper is structured as follows. In Materials and Methods, the problem framework is outlined

and, subsequently, the stochastic modeling approach is presented. The numerical studies and most relevant findings are discussed in Results and Discussion. Finally, the concluding remarks and an outlook on future research opportunities are provided in the Conclusion section.

Materials and Methods

Problem framework and Modeling approach

The interplanetary network

The directed network structure consists of several nodes and connecting arcs along which cargo can be transported. These nodes correspond to orbits and celestial bodies, which are potentially relevant for interplanetary travel. Notably, the considered ISC network actually consists of 2 interacting networks, i.e., a primary and a secondary network (PN and SN, respectively). The PN is concerned with delivering cargo from Earth to Mars, while the SN is responsible for the propellant supply along the way. Both of these share the same nodes but comprise different arcs based on their distinct purposes. This results in a layered network where the nodes represent points of interactions between the 2 layers. Figure 1 provides a schematic view of the layered network concept.

Figure 2 shows the complete PN structure considered in the developed model. It is based on space logistics networks that have been used for Mars mission planning in the extant scientific literature (e.g., [4,12,15]) and in the MIT's Space Logistics Project [7]. Notably, the network includes several surface nodes, i.e., planets (Earth and Mars) and their satellites (Moon, Phobos, and Deimos), numerous orbital nodes (low Earth orbit (LEO), geostationary orbit (GEO), low lunar orbit (LLO), Lagrange points (L1, L2, and L4/5), low Mars orbit (LMO), low Phobos orbit (LPO), and low Deimos orbit (LDO)), and 2 auxiliary nodes (lunar transfer orbit (LTO) and Deimos transfer orbit (DTO)). Every one of these nodes is either part of the Earth cluster or the Mars cluster. Routes within the same cluster are not time variant and are thus available at all times. On the other hand, traveling from one cluster to another (i.e., from LTO to DTO) is not always possible and depends on the current planetary constellations. The characteristics of the corresponding interplanetary transfer windows are discussed in detail in a subsequent section. Notably, the PN has no connecting arcs to the Moon, Phobos, or Deimos because layover stops at these satellites would require too much energy.

The SN, shown in Fig. 3, is concerned with meeting the propellant demands that arise from the transportation processes in the PN. Propellant can be manufactured on surface nodes and is supplied to spacecraft on Earth and, if refueling is necessary, in space. To store propellant in space, on-orbit depot have to be set up at the respective node. To deliver propellant to the said on-orbit depots, the SN comprises arcs from surface nodes to orbital nodes within the same cluster (excluding the 2 auxiliary nodes). However, not all possible arcs are available as the propellant requirements for fuel transportation processes along some routes are prohibitively high. The feasibility of each supply route within the SN is determined by means of a short optimization program detailed in Section S4. Intercluster arcs are not considered because propellant supply over such distance is too inefficient. All nodes where spacecraft receive propellant represent points of interaction between the 2 network layers.

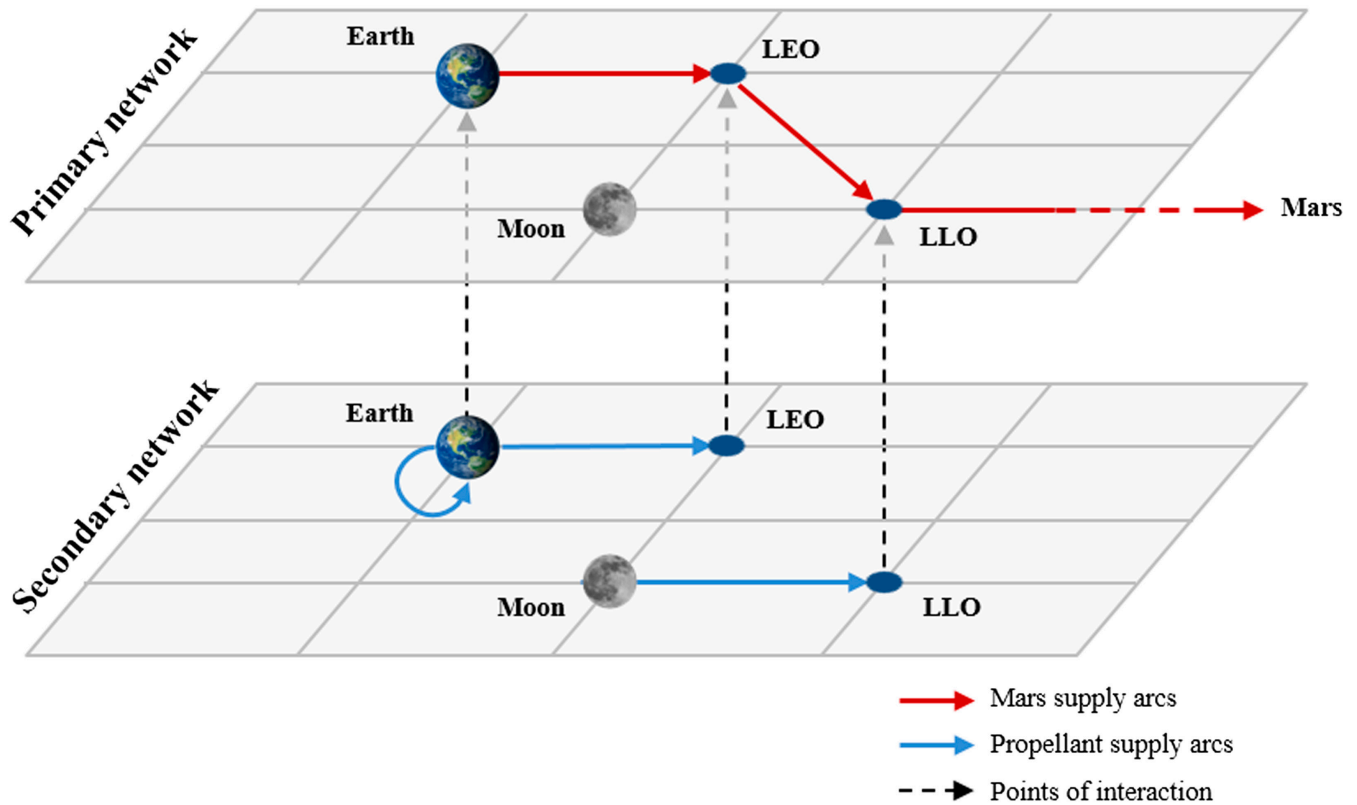


Fig.1. Schematic view of the layered network structure. LEO, low Earth orbit; LLO, low lunar orbit.

Surface nodes: These nodes represent celestial bodies and generally serve as locations where supplies and demands are generated. Notably, Earth and Mars have unique characteristics in that the latter generates demands, which have to be met with materials exclusively supplied from the former. All surface nodes, however, may be used for propellant production, which requires the installation of ISRU plants on site. Earth is the only exception as propellant can be provided through traditional channels without setting up ISRU facilities. It is noteworthy that the planetary satellites (i.e., Moon, Phobos, and Deimos) exclusively serve as potential locations for ISRU plants and are, thus, not accessible through the PN.

Orbital nodes: The considered orbital nodes correspond to relatively stable orbits that are accessible from the aforementioned surface nodes. They represent possible locations for on-orbit depots and, thus, for the refueling en route to Mars. Notably, the network structure requires spacecraft to always go through the nodes LEO and LMO when leaving Earth or descending on Mars, respectively. This is a modeling adjustment that has been adopted from the scientific literature to reduce the number of arcs in the networks and, thus, to keep the model at a manageable size (e.g., [4,12,15]). Traveling through an orbital node does not necessarily mean that a spacecraft spends much time in this orbit but may merely be passing through. In fact, layovers may only be sensible for refueling purposes. Some orbital nodes, i.e., the Lagrange points of the Earth–Moon system, are of special importance and are, thus, discussed in more detail in the following section.

Lagrange points: Lagrange points play an important role as potential positions for on-orbit infrastructure as they are

points of enhanced stability. Therefore, deployed structures, such as space stations, require little to no station-keeping in order to maintain their position over time. This is a very desirable state and relevant for countless space applications. All 5 Lagrangian points theoretically provide suitable positions for on-orbit infrastructure. L3, however, can be omitted in the considered network as the other points are preferred for in-space infrastructure because of their closeness to Earth and Moon (L1, L2) and their increased stability (L4, L5). In the model, L4 and L5 are represented by a single node (L4/5) because their locations are mirrored along the Earth–Moon axis and, thus, identical efforts are required to reach them.

Auxiliary nodes: LTO and DTO are considered as auxiliary nodes, which, together, serve as the gateway for intercluster travel. In the proposed network, LTO always serves as the staging node for trans-Mars injection. DTO, on the other hand, is the destination of all interplanetary trips. Very similar approaches have been adopted by, for instance, Ho [13] and Chen et al. [16] who also consider the arc from LTO to DTO to be the only connection between the Earth and Mars clusters. The usage of auxiliary nodes as the only intercluster gateway helps to reduce the number of arcs and, thus, the network complexity. Moreover, it provides an efficient way to implement interplanetary transfer windows because only one arc is affected by them. In accordance with Ho [13], these transfer orbits cannot hold any permanent structures such as on-orbit depots.

Useful transfer orbits between Earth's and Mars' orbits are not available at all times, but only during the so-called transfer window. A transfer window describes a point in time at which an intercluster transfer is feasible. Otherwise, a transfer between

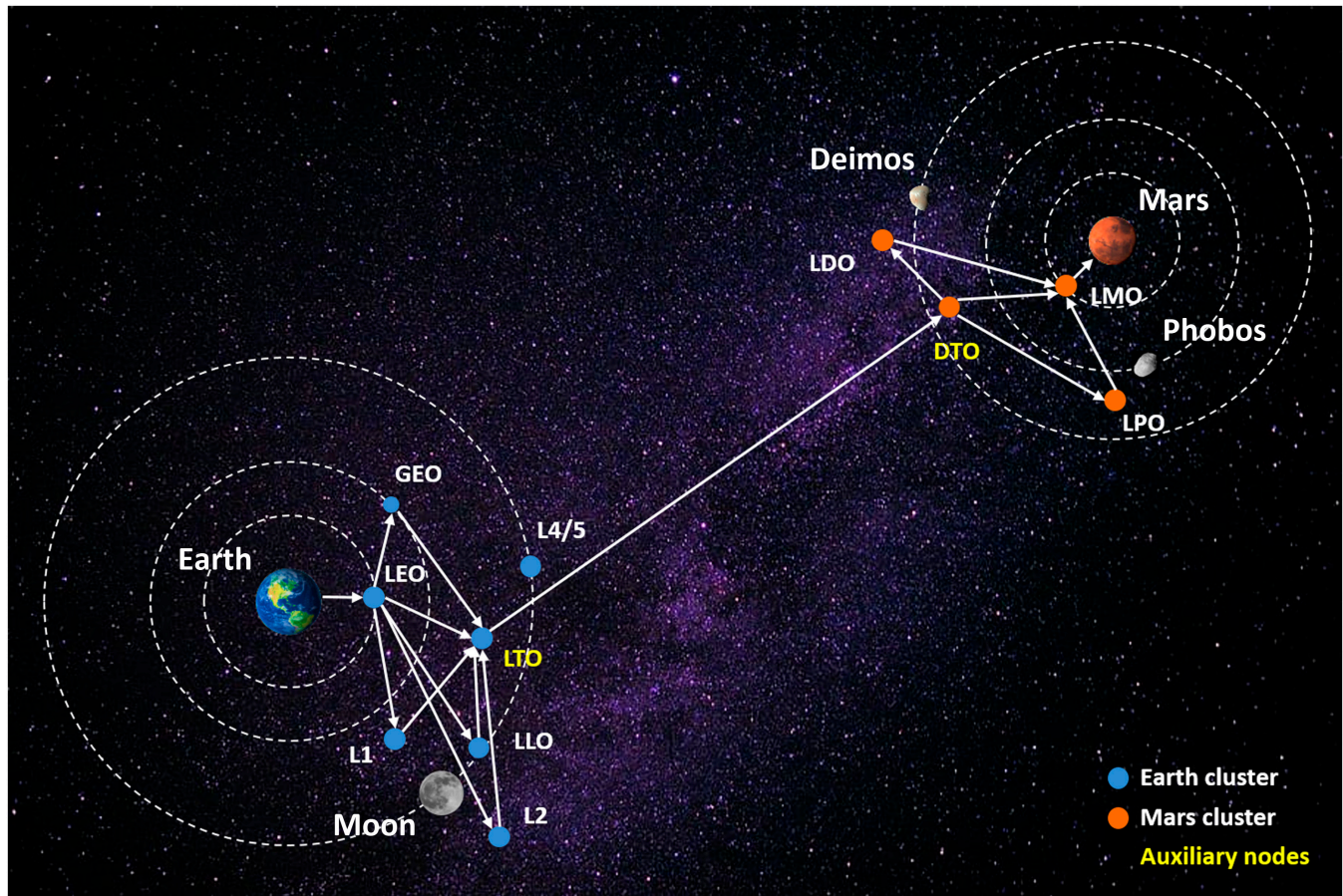


Fig. 2. Primary network (PN) structure to supply martian base. LDO, low Deimos orbit; LMO, low Mars orbit; DTO, low Phobos orbit; LPO, low Phobos orbit; GEO, geostationary orbit; LTO, lunar transfer orbit.

Earth and Mars is associated with prohibitively high propellant requirements. In other words, the availability of routes and the corresponding times of flights (TOF) and costs (i.e., propellant requirements) are time dependent. Moreover, within the same transfer window, there might be several trajectories available with different characteristics. Typically, there is a trade-off between TOF and cost that decision-makers have to balance when choosing a trajectory.

Propellant consumption

The transportation of cargo through space, let alone to another planet, is an extremely energy-intensive endeavor. Hence, propellant is a critical asset in space travel that has to be accounted for as a limiting element in the model. Propellant consumption essentially depends on 3 factors: (a) the propellant type used, (b) the desired change in velocity, and (c) the total mass being moved.

(a) Various types of propellant can be used in chemical propulsion systems such as LOX/LH₂ and LOX/LCH₄. Spacecraft typically have 2 tanks, one for the oxidizing agent (liquid oxidizer or LOX) and one for the actual fuel (e.g., liquid hydrogen (LH₂) or liquid methane (LCH₄)). The crucial difference between the various propellant types lies in their specific impulses, I_{sp} . The specific impulse essentially describes how much thrust is generated by a standardized amount of propellant.

(b) The change in velocity required to travel from one orbit to another is generally quantified as Δv . Hence, a specific Δv exists for each considered arc in the model. For all arcs within the same cluster (i.e., Earth or Mars cluster), Δv values can be

considered a constant, e.g., traveling from LEO to GEO. However, the Δv for trans-Mars injection changes over time as it depends on the relative positions of Earth and Mars. In other words, the fuel requirements for intercluster flights are time dependent. Moreover, faster and slower transit trajectories may exist at the same point in time requiring more or less fuel, respectively. Hence, the model has to manage a trade-off between fuel requirements and TOF.

(c) Three different mass parameters are relevant for the propellant consumption of a spacecraft, i.e., its wet mass (m^{wet}), its dry mass (m^{dry}), and the propellant mass (m^p). The relationship between the 3 is as follows: $m^{\text{wet}} = m^{\text{dry}} + m^p$. The dry mass refers to the combined masses of the vehicle's physical structure and cargo. The wet mass, on the other hand, is equal to the sum of the dry mass and propellant mass.

When all these parameters are established, the actual propellant requirements for a given arc (i, j) can be computed. This requires the calculation of the corresponding propellant mass fraction, $\phi_{(i,j)}$ (Eq. 1), which is directly derived from the rocket equation [4,12]. The propellant mass fraction determines the fraction of the wet mass that does not reach the destination, i.e., the burned fuel. In other words, it says what share of the wet mass is required to be propellant in order to move the remaining share from i to j . Note that the remaining share includes the dry mass and any additional fuel loaded. The propellant mass fraction takes as input the change in velocity (Δv) required for the trip, the specific impulse of the chosen fuel type (I_{sp}), and the standard gravity (g_0).

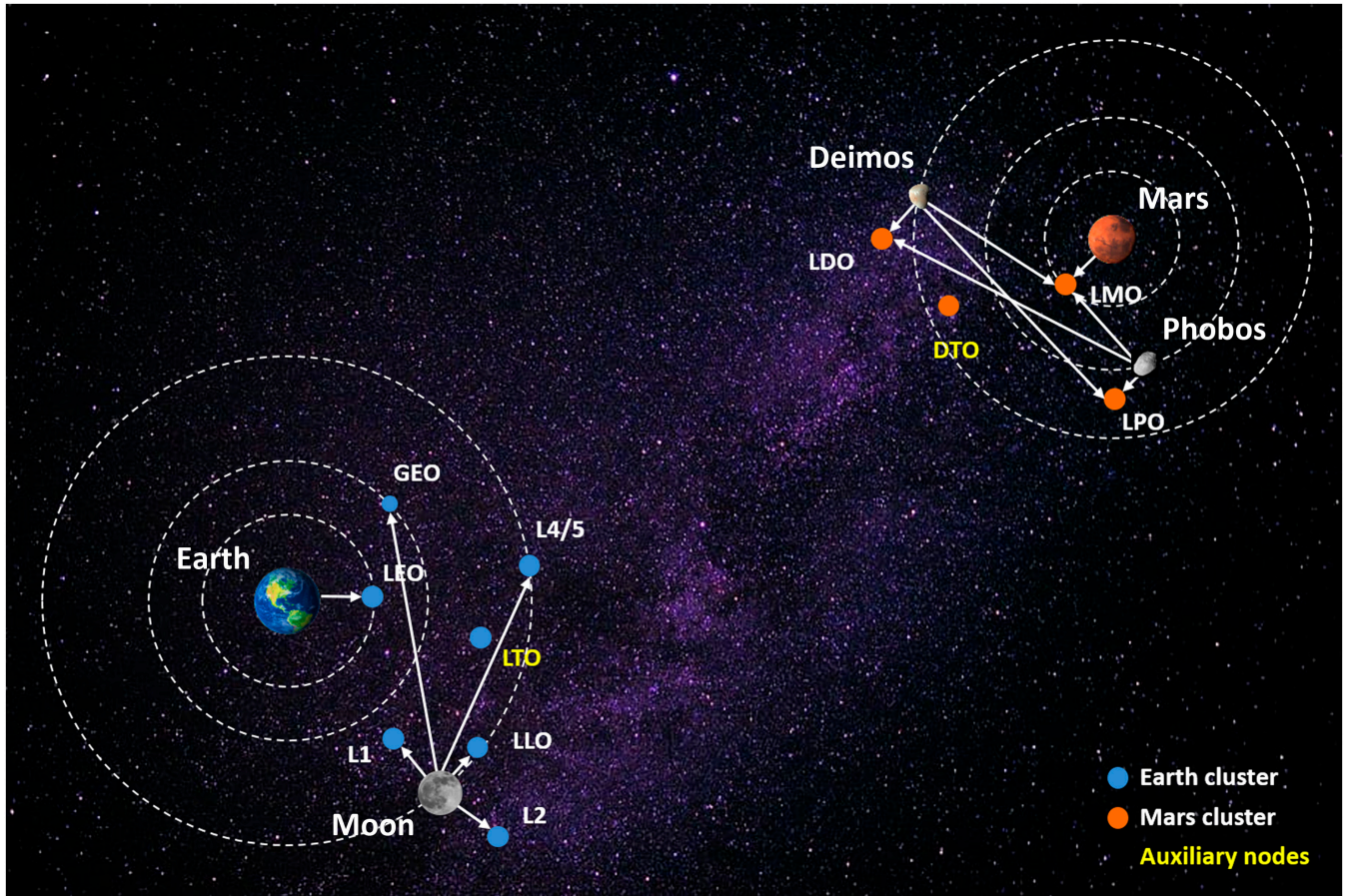


Fig. 3. Secondary network (SN) structure for propellant supply.

$$\phi_{(i,j)} = 1 - \exp\left(-\frac{\Delta v_{(i,j)}}{I_{sp} \cdot g_0}\right) \quad (1)$$

The resulting $\phi_{(i,j)}$ can be used to calculate the propellant consumption for a given arc (i,j) as shown in Eq.2 [4,12]. The equation takes into account the dry mass of the vehicle (m^{dry}), the fuel mass at the beginning of the trip (m_{start}^p), and the factors in the propellant mass fraction ($\phi_{(i,j)}$). The result equates to the amount of propellant left at the end of the trip (m_{end}^p). Therefore, the difference between m_{start}^p and m_{end}^p is the amount of fuel burned. This equation plays a key role in the upcoming optimization model as it determines the propellant requirements for each individual arc (i,j) . Note that, throughout the remainder of this manuscript, all mass values refer to metric tons [t].

$$m_{\text{end}}^p = -\phi_{(i,j)} \cdot m^{\text{dry}} + (1 - \phi_{(i,j)}) \cdot m_{\text{start}}^p \quad (2)$$

Modeling Approach

Multistage stochastic programming

The concept of uncertainty reflects the reality that many parameters in decision problems suffer from incomplete information. It follows that decisions have to be made without knowing exactly how parameter values are going to develop in the future. Stochastic programming, introduced by Dantzig [17], is an

established approach to account for such uncertainty in optimization problems. Here, uncertainty is represented through random variables, which follow specified probability distributions. Moreover, the optimization approach requires to determine a discrete set of scenarios whereby each scenario captures a distinct realization of the random variables. Therefore, the scenario set essentially constitutes a discrete approximation of the original probability distribution.

In 2-stage stochastic programming, uncertainty only exists up to a specific moment in time. After this point, the uncertainty is resolved and the model essentially consists of multiple deterministic scenarios for all subsequent periods (scenario fan). MSP, on the other hand, is able to reflect uncertainty at multiple points in time, which leads to a so-called scenario tree [18]. This characteristic fits particularly well with the periodic transfer windows for interplanetary space flights. The transfer windows determine specific points in time when decisions under uncertainty have to be made.

The proposed MSP model for ISC planning considers demands at a martian base as a source of uncertainty. Given that Earth–Mars transfer times can be extremely long, demand uncertainty can be expected to play a major role in interplanetary logistics. As a consequence, 2 classes of decision variables are differentiated in the developed MSP model: There are strategic scenario-independent variables, which represent decisions that have to be made before the uncertainty unfolds. These decisions cannot be altered over the course of the planning horizon as this would be too costly. Tactical scenario-dependent

variables, on the other hand, are made after the uncertain parameters have materialized and can be adjusted over time. Both of these variable classes, however, are subject to uncertainty. The allocation decisions regarding on-orbit depots (Dep_i) and ISRU facilities (ISRU_i) belong to the former class as the installation of such structures can be considered an extremely costly and time-consuming endeavor. Depot and ISRU facility sizes are reflected by continuous variables. Hence, these infrastructure elements are considered to be linearly scalable, which is a common assumption in space logistics (e.g., [11,16]).

The tactical scenario-dependent decisions include the number of spacecraft ($R_{(i,j,t,t'),s}$), dry masses ($D_{(i,j,t,t'),s}$), payloads ($Q_{(i,j,t,t'),s}$), and propellant volumes at the departure ($P_{(i,j,t,t'),s}^{\text{depart}}$) and arrival ($P_{(i,j,t,t'),s}^{\text{arr}}$) nodes of arc (i,j) . All the aforementioned decisions have to be made for transfer windows (i,j,t,t') . This set consists of a departure and an arrival node (i and j , respectively) as well as a departure and arrival period (t and t' , respectively). For all trips within the same cluster (Earth–Moon cluster or martian cluster), the arrival periods are equal to the departure periods ($t = t'$) because of the relative closeness of the nodes. This includes propellant supplies from ISRU facilities ($P_{(h,i),t,s}^{\text{ISRU}}$), which are restricted to intra-cluster routes. On the other hand, all trips between both clusters (interplanetary trips) have a much longer travel time, which results in $t < t'$. The remaining scenario-dependent decisions (propellant inventories at depots ($I_{i,t,s}^{\text{Dep}}$), spacecraft inventories ($I_{i,t,s}^R$), payloads ($I_{i,t,s}^Q$), and propellant supplies ($P_{i,t,s}^{\text{sup}}$)) are made for individual nodes i .

Notably, this model addresses the transportation of materials (spacecraft and cargo to Mars and propellant to spacecraft) as is typical in SC optimization. The transportation of people, i.e., astronauts, to Mars is omitted because it has very different requirements. Hence, the manned missions are assumed to be planned separately and, thus, are out of the scope of the developed model.

Model notations

All sets of indices, parameters, and variables used in the subsequent model are listed below.

Index sets

N	Index set of all investigated nodes
$N^{\text{Space}} \subseteq N$	Set of orbital nodes
$N^{\text{Aux}} \subseteq N$	Set of auxiliary nodes
N^{Surf}	Set of surface nodes excluding Earth
X^{sup}	Set of arcs (h,i) indicating that propellant was produced at $h \in N^{\text{Surf}}$ and supplied at $i \in N$
S	Set of scenarios
$S_k \subseteq S$	Sets $k = 1 \dots K$ of scenarios on stage 2
W	Set of transfer trajectory (i,j,t,t') from location $i \in N$ to $j \in N$, departing in period $t = 1 \dots T$ and arriving in $t' = 1 \dots T$

Parameters

m	Structural mass of a spacecraft
α	Structural mass fraction for ISRU plants
β	Structural mass fraction for on-orbit depots
γ_i	Mass fraction for allocating infrastructure to node i
$\delta_{(h,i)}$	Mass fraction for propellant supply given $(h,i) \in X^{\text{sup}}$
cap_i^p	Propellant capacity of spacecraft departing from node i
cap_i^Q	Payload capacity of spacecraft
$d_{i,t,s}$	Demand at node i in period t of scenario $s \in S$
$\phi_{(i,j,t,t')}$	Propellant mass fraction for transfer window $(i,j,t,t') \in W$
prob_s	Probability of scenario s
T	Planning horizon
K	Number of scenario subsets on the second stage 2
t^a	Final period of stage 1
t^b	Final period of stage 2

Strategic decision variables

$\text{Dep}_i \in \mathbb{R}_0^+$	Size of the depot at $i \in N^{\text{Space}}$
$\text{ISRU}_i \in \mathbb{R}_0^+$	Size of ISRU plant established at $i \in N^{\text{Surf}}$

Tactical decision variables

$I_{i,t,s}^{\text{Dep}} \in \mathbb{R}_0^+$	On-orbit depot propellant inventory at node $i \in N$ in period $t = 1 \dots T$ of scenario $s \in S$
$I_{i,t,s}^Q \in \mathbb{R}_0^+$	Payload inventory at node $i \in N$ in period $t = 1 \dots T$ of scenario $s \in S$
$I_{i,t,s}^R \in \mathbb{Z}_0^+$	Number of spacecraft parked at node $i \in N$ in period $t = 1 \dots T$
$R_{(i,j,t,t'),s} \in \mathbb{Z}_0^+$	Spacecraft traveling along transfer window $(i,j,t,t') \in W$
$Q_{(i,j,t,t'),s} \in \mathbb{R}_0^+$	Payload carried along transfer window $(i,j,t,t') \in W$ in scenario $s \in S$
$P_{(i,j,t,t'),s}^{\text{depart}} \in \mathbb{R}_0^+$	Propellant at departure for transfer window $(i,j,t,t') \in W$ and scenario $s \in S$
$P_{(i,j,t,t'),s}^{\text{arr}} \in \mathbb{R}_0^+$	Propellant at arrival for transfer window $(i,j,t,t') \in W$ and scenario $s \in S$
$P_{i,t,s}^{\text{sup}} \in \mathbb{R}_0^+$	Propellant supply to spacecraft at $i \in N$ in period $t = 1 \dots T$ of scenario $s \in S$
$P_{(h,i),t,s}^{\text{ISRU}} \in \mathbb{R}_0^+$	Propellant generated and supplied from ISRU facility to in-space node through arc $(h,i) \in X^{\text{sup}}$ in period $t = 1 \dots T$ of scenario $s \in S$
$D_{(i,j,t,t'),s} \in \mathbb{R}_0^+$	Dry mass of spacecraft for transfer window $(i,j,t,t') \in W$ and scenario $s \in S$

Model formulation

On the basis of the previous model description, the following multistage stochastic mixed integer linear programming (MILP) model is developed.

$$\min \sum_{i \in N^{\text{Space}}} \text{Dep}_i \cdot \gamma_i + \sum_{i \in N^{\text{Surf}}} \text{ISRU}_i \cdot \gamma_i + \sum_{t=1 \dots T} \sum_{t'=1 \dots T} \text{prob}_s \cdot \left(\sum_{(i,j,t,t') \in W: i=\{\text{Earth}\}} D_{(i,j,t,t'),s} + \sum_{(h,i) \in X^{\text{sup}}} P_{(h,i),t,s}^{\text{sup}} \cdot \delta_{(h,i)} \right) \quad (3)$$

The objective function shown in Eq. 3 minimizes the total launch mass (TLM), which is based on the premise that every asset utilized in space must have been launched from a surface node at some point in time. This includes on-orbit depots and ISRU facilities. Moreover, it comprises the total dry mass (spacecraft + payload) launched from Earth throughout the planning horizon and the propellant supplied at any point of the network. Note that propellant is the only asset that is not required to be launched from Earth because ISRU technology enables production on other celestial bodies (i.e., Moon, Mars, Phobos, and Deimos).

$$I_{i,t,s}^R = I_{i,t-1,s}^R + \sum_{(j,i,t',t) \in W} R_{(j,i,t',t),s} - \sum_{(i,j,t,t') \in W} R_{(i,j,t,t'),s} \quad \forall i \in N, s \in S; t = 1 \dots T \quad (4)$$

$$I_{i,t,s}^Q = I_{i,t-1,s}^Q + \sum_{(j,i,t',t) \in W} Q_{(j,i,t',t),s} - \sum_{(i,j,t,t') \in W} Q_{(i,j,t,t'),s} - d_{i,t,s} \quad \forall i \in N, s \in S; t = 1 \dots T \quad (5)$$

$$\sum_{(j,i,t',t) \in W} P_{(j,i,t',t),s}^{\text{arr}} + P_{i,t,s}^{\text{sup}} = \sum_{(i,j,t,t') \in W} P_{(i,j,t,t'),s}^{\text{depart}} \quad \forall i \in N, s \in S; t = 1 \dots T \quad (6)$$

$$I_{i,t,s}^{\text{Dep}} = I_{i,t-1,s}^{\text{Dep}} + \sum_{(h,i) \in X^{\text{sup}}} P_{(h,i),t,s}^{\text{ISRU}} - P_{i,t,s}^{\text{sup}} \quad \forall i \in N, s \in S, t = 1 \dots T \quad (7)$$

Equations 4 to 6 control the flows of spacecraft, payload, and propellant in the PN, while Eq. 7 is concerned with the propellant flows in the SN. The demand parameters in Eq. 5 have positive values ($d_{j,t,s} > 0$) at demand points (i.e., Mars) and equal 0 at all other nodes. It is noteworthy that propellant inventories are not accounted for in the PN because this resource is completely consumed at the end of the journey. Carrying along any additional quantities would be inefficient and, thus, avoided by the model in any case. In the SN, however, propellant inventories are accounted for because fuel can be stored across periods at on-orbit depots (see Eq. 7).

$$P_{(i,j,t,t'),s}^{\text{arr}} = -\phi_{(i,j,t,t'),s} \cdot D_{(i,j,t,t'),s} + (1 - \phi_{(i,j,t,t'),s}) \cdot P_{(i,j,t,t'),s}^{\text{depart}} \quad \forall (i,j,t,t') \in W, s \in S \quad (8)$$

$$m \cdot R_{(i,j,t,t'),s} + Q_{(i,j,t,t'),s} \leq D_{(i,j,t,t'),s} \quad \forall (i,j,t,t') \in W, s \in S \quad (9)$$

Equation 8 calculates the amount of fuel left over after traveling a specific arc. Because the amount of propellant on each spacecraft is capacitated, this equation implicitly constraints the payload mass. Equation 9 provides the dry mass calculation based on the structural and payload masses.

$$P_{(i,j,t,t'),s}^{\text{depart}} \leq R_{(i,j,t,t'),s} \cdot \text{cap}_i^P \quad \forall (i,j,t,t') \in W, s \in S \quad (10)$$

$$Q_{(i,j,t,t'),s} \leq R_{(i,j,t,t'),s} \cdot \text{cap}_i^Q \quad \forall (i,j,t,t') \in W, s \in S \quad (11)$$

Equations 10 and 11 are capacity constraints for a fleet of spacecraft $R_{(i,j,t,t'),s}$. The propellant capacity is location dependent because spacecraft bound for Mars would be launched with staged rockets from Earth. Thus, they have a much greater propellant capacity for the trip from Earth to LEO and a lower capacity for all subsequent arcs.

$$I_{i,t,s}^Q \leq I_{i,t,s}^R \cdot \text{cap}_i^Q \quad \forall i \in N^{\text{Space}}, s \in S; t = 1 \dots T \quad (12)$$

$$I_{i,t,s}^R \leq 0 \quad \forall i \in N^{\text{Space}}, s \in S; t = 1 \dots T \quad (13)$$

$$I_{i,t,s}^{\text{Dep}} \cdot \beta \leq \text{Dep}_i \quad \forall i \in N^{\text{Space}}, s \in S; t = 1 \dots T \quad (14)$$

$$\text{Dep}_i \leq 0 \quad \forall i \in N^{\text{Aux}}, s \in S; t = 1 \dots T \quad (15)$$

$$\sum_{(h,i) \in X^{\text{sup}}} P_{(h,i),t,s}^{\text{ISRU}} \cdot \alpha \leq \text{ISRU}_h \quad \forall h \in N^{\text{Surf}}, s \in S; t = 1 \dots T \quad (16)$$

Equations 12 and 13 in combination restrict the amounts of payload and spacecraft inventories to 0 for all locations aside from Earth and Mars. This means that spacecraft travels to Mars without multiperiod stopovers at intermediate nodes. Furthermore, Eq. 14 dictates that propellant supplies presuppose the installation of depots at the respective locations with the exception of Earth. Such depots, however, cannot be established at auxiliary nodes (Eq. 15). Moreover, propellant supplies, provided from any surface node but Earth, require the installation of ISRU facilities at location h to manufacture said propellant (Eq. 16).

$$I_{i,t,s}^Q = 0 \quad \forall i \in N \setminus \{\text{Earth}\}, s \in S, t = 0 \quad (17)$$

$$I_{i,t,s}^R = 0 \quad \forall i \in N \setminus \{\text{Earth}\}, s \in S, t = 0 \quad (18)$$

$$I_{i,t,s}^{\text{Dep}} = 0 \quad \forall i \in N, s \in S, t = 0 \quad (19)$$

Equations 17 and 18 initialize payload and spacecraft inventories to 0 for all nodes except Earth. Equation 19 achieves the same for propellant inventories at on-orbit depots.

$$Q_{(i,j,t,t'),s} = Q_{(i,j,t,t'),s'} \quad \forall (i,j,t,t') \in W: t \leq t^a, s, s' \in S: s \neq s' \quad (20)$$

$$Q_{(i,j,t,t'),s} = Q_{(i,j,t,t'),s'} \quad \forall (i,j,t,t') \in W: t^a < t \leq t^b, s, s' \in S_k: s \neq s', k \in K \quad (21)$$

Equations 20 and 21 enforce non-anticipativity for payload transportation decisions over 3 stages. For stage 1, payload transportation decisions have to be the same in every scenario. On stage 2, payloads only have to be equal for all scenarios of the same scenario subset S_k . Through Eqs. 6 to 11, the non-anticipativity implicitly extends to the number of spacecraft, dry masses, propellant consumption, and propellant supplies (i.e., all transportation decisions).

Results and Discussion

Outline of the case example

The proposed model is applied to a case example concerned with consistently supplying a permanent base on Mars. The considered planning horizon comprises 6 years (on Earth) divided in 24 periods of 3 months each. This long time frame is chosen because of the strategic planning level and to limit end

of horizon effects. Moreover, 6 years on Earth roughly equate to 3 martian synodic periods. A synodic period equals about 780 d for Mars, which roughly translates to 2 years on Earth and 8 periods in the proposed model. The duration of the synodic period provides the basis for the different stages of the optimization model. Accordingly, the first stage ranges from $t = 1 \dots 8$ (therefore, $t^a = 8$), the second stage from $t = 9 \dots 16$ (therefore, $t^b = 16$), and the third stage from $t = 17 \dots 24$.

Periodic demands have to be fulfilled using the prespecified network presented in The interplanetary network. The Δv values for all intracluster arcs, which are used to calculate the corresponding propellant consumption, are drawn from the literature [11] and can be viewed in Table S2. Because of the chosen period length, flights within the same cluster, which only take a few days, are assumed to reach their destination immediately.

The Δv values and TOFs for intercluster flights are time variant and depend on the specific transfer trajectory. To find suitable transfer windows, the trajectory browser developed by the NASA Ames Research Center was used. This online tool identifies possible transfer trajectories from Earth, more specifically LEO, to other planets and small bodies within the solar system. It returns relevant information on each trajectory such as the launch date, Δv requirements, and the TOF. A more detailed description of its functionalities is provided by Foster [19].

The trajectory browser requires the user to set some mission constraints, which are detailed in Section S1. The query yielded 32 different trajectories in total. The departure and arrival dates of these trajectories were then translated into the equivalent periods that are used in the model (1 period = 3 months). The first period, therefore, corresponds to the first 3 months of the year 2030 (January, February, and March), the second period to the second 3 months (April, May, and June), and so on until the 24th period. In a subsequent preprocessing step, all trajectories that are dominated by another were eliminated from the set. A trajectory is considered to be dominated if another trajectory exists that has identical departure and arrival periods but lower Δv requirements. The final set of transfer windows comprised 12 different trajectories (Table S3). As mentioned before, the trajectory browser provides the Δv requirements when departing from LEO. In the proposed PN, however, trans-Mars injection is carried out at the auxiliary node LTO in accordance with relevant literature in this field [13,16]. It follows that the Δv requirements returned by the trajectory browser have to be adjusted for the differing locations. To this end, the Δv required to fly from LEO to LTO (3.18 km/s) is subtracted from the values provided by the trajectory browser. The resulting values correspond to the Δv necessary to travel from LTO to DTO. In total, the route LEO $\rightarrow$ LTO $\rightarrow$ DTO, thus, retains the original Δv requirements provided by the trajectory browser.

For brevity, most of the parameter calculations and remaining parameter settings are presented in Sections S2 and S3, respectively. The exception being demand and scenario generation processes, which are discussed in the following.

Demand generation

A seasonal demand pattern is established as the baseline case. The assumption is that demands are affected by the quite harsh seasonal conditions on Mars. Moreover, a year on Mars is about twice as long as an Earth year and the same is true for its seasons. Note that a martian year refers to a complete revolution around the sun regardless of the positioning of Earth. As a result, the

martian year is not identical with its synodic period as it is slightly shorter. However, given the considered period length and granularity of the planning model, a distinction between the 2 is not necessary and, hence, 2 years are assumed for both. On the basis of the approach used by Hahn and Kuhn [20], seasonal demands for each period and scenario are calculated as shown in Eq. 22.

$$d_{i,t,s} = f_{i,s} \cdot d_i^{\text{base}} \cdot \left(1 + \text{amp} \cdot \sin \left(\frac{2 \cdot \pi}{T^c} \left(t + \frac{T^c - 1}{2} \right) \right) \right) \forall i = \{\text{Mars}\}, s \in S, t = 11 \dots T \quad (22)$$

Given the long transportation times, an immediate supply of the martian base at the beginning of the planning horizon would be impossible. Therefore, it is established that demands do not occur before period 9, which provides sufficient opportunity to deliver the supplies in time. The scenario factor $f_{i,s}$ will be used to generate the desired number of scenarios from the baseline case.

The parameter d_i^{base} represents the base level of demand and is set to 0 for all nodes i except for Mars. The demand base level for Mars is determined from the data provided in NASA's Mars Design Reference Architecture 5.0 [21]. In the Design Reference Architecture 5.0, a provision mass of 16.793 t is required to support a crew of 6 for 1,000 d in the Mars Transit Habitat. Given that one period equals to 91.25 d, this translates to roughly 0.255 t of provisions per individual crew member and period. Provisions, thereby, comprise all supplies the crew might need during their stay, which include food, healthcare products, maintenance equipment, spare parts, and other things of that nature. For this case study, a population of 100 people is assumed, resulting in a demand of 25.5 t per period. However, a martian base can be expected to have higher demands than a Mars Transit Habitat because of the multitude of additional activities that are carried out. This may include materials and equipment to expand the base, to maintain ISRU facilities, and to promote research efforts. Thus, the base level of demand is estimated at $d_i^{\text{base}} = 50$ t for $i = \text{Mars}$.

The remaining term of Eq. 22 generates the seasonal demand pattern from the demand base level by assuming a harmonic oscillation with an amplitude of $\text{amp} = 0.1$. Moreover, the length of the seasonal cycle is estimated at $T^c = 8$ periods, which translates to 1 Mars year. The seasonal peak corresponds to the martian winter, which comprises 2 periods as it is twice as long as on Earth.

Scenario generation

Demands are considered to be uncertain in every scenario and period and are modeled as random variables accordingly. Moreover, it is assumed that demands follow a normal distribution, whereas the previously established baseline demand pattern constitutes the expected values. The degree of uncertainty is determined by the variation coefficient, which is estimated at $\sigma/\mu = 0.1$. This implies a relatively small σ value compared to the ones often used in models of terrestrial SCs. Terrestrial SCs obviously assume an existing market economy where demand uncertainty is a result of a multitude of factors including consumer behavior and competition. It follows that a greater degree of uncertainty that needs to be considered in the respective models. However, a martian base would be an extremely controlled environment where demands mainly arise from consumption, expansion, and maintenance. The corresponding

uncertainty simply reflects variations in these activities because of unforeseen developments or events.

To account for uncertainty in MSP models, however, a discrete set of scenarios needs to be generated from the described normal distribution. Each of these scenarios comprises discrete realizations of every uncertain parameter [22], which is captured in the scenario factor $f_{i,s}$ (see Eq. 22). To achieve the discretization of the random variables, the moment-matching scenario generation heuristic proposed by Høyland et al. [23] is applied. This approach samples the desired number of scenarios from a prespecified continuous distribution, i.e., a normal distribution in this case. Importantly, it manages to preserve the first 4 moments of that distribution within the generated scenario set. Therefore, these scenarios retain quintessential characteristics of the original probability function. This reduces the amount of scenarios required to accurately represent the considered uncertainty in the model.

There is a trade-off between model accuracy and tractability when it comes to scenario set size. The more scenarios are considered, the more accurately they reflect the true probability distribution of the stochastic event. The computation time, on the other hand, may increase exponentially with the number of scenarios [24], which may quickly render the model intractable. To determine a scenario set size that balances this trade-off, the in-sample stability test proposed by Kaut and Wallace [25] is conducted. In-sample stability is achieved when different but equally sized scenario sets, which were generated with the same method, yield similar objective function values (OFVs). This ensures that the results of the optimization model are not driven by the currently used sample. For this test, various scenario set sizes are investigated by generating 20 different instances each. We find that 25 scenarios (5 on the second stage and 5 on the third stage) provide sufficient in-sample stability because the standard deviation of the OFVs is less than 0.15% of the average OFV. This high degree of stability is achieved with a relatively small scenario set because, once again, demand uncertainty in such a controlled environment is not as high as in terrestrial SCs. However, omitting this stochasticity altogether would bear the risk of stock-out situations, which would have dramatic consequences for the martian base and, thus, needs to be avoided categorically.

The stochastic MILP model is implemented in IBM ILOG CPLEX Optimization Studio v20.1.0. It comprises 564,726 constraints, 19,225 integer variables, and 76,268 continuous variables. The optimization problem is solved to optimality on a Windows PC with an Intel Core i7-8700U CPU @ 3.20 GHz and 16-GB RAM in an average run time of 130 s.

Baseline case

The results of the baseline case with regards to the total mass launched can be viewed in Table 1. It shows that the installation of the space infrastructure is responsible for more than 37% of the TLM, with ISRU facilities and on-orbit depots sharing roughly equal parts. Despite their similar masses, the storage capacity of on-orbit depots is much higher than the production capacity of ISRU plants because of the differences in the respective structural mass fractions. Notably, the expected dry mass, which includes the structural weight and all delivered cargo, only comprises 0.36% of the TLM. The expected propellant mass amounts to almost 62% and thus makes up the majority of the TLM. It follows that roughly 170 times more propellant than dry mass is required to support the martian base over the planning horizon.

Table 1. Breakdown of total launch mass (TLM) in baseline case.

	Launch mass [t]	Share of TLM
ISRU facilities	88,143	20.26%
Depots	75,985	17.47%
Exp. dry mass	1,565	0.36%
Exp. propellant supply	269,325	61.91%
TLM	435,018	100%

The complete routing through the SC network (PN and SN) and the allocation of in-space infrastructure are shown in Fig. 4. ISRU facilities are set up on all planetary satellites, i.e., Moon, Phobos, and Deimos, while depots are established at L2, LDO, and LMO. Interestingly, not all spacecraft follow the same routing through the PN. In the Earth cluster, a fleet of spacecraft often split up to pursue a division-of-labor approach. In this case, 3 of 4 launched spacecraft fly directly from LEO to LTO, carrying most of the total payload (cargo ships). The fourth spacecraft, on the other hand, is transporting less cargo and serves the purpose of a propellant tanker. It takes a longer route, traveling first to L2 in order to load up propellant at the corresponding depot. Then, it rejoins the other spacecraft at LTO and supplies them with fuel. Thus, the tasks are somewhat divided as some vehicles are responsible for the bulk of the cargo while others fetch propellant from the depot. Subsequently, all 4 spacecraft engage in trans-Mars injection but do not follow the same trajectories. The cargo ships take the slower but more propellant-efficient transfer window VI, while the tanker takes the quicker but more-propellant intensive window V. Both transfer windows require departing in period 4 but have different TOFs, causing the tanker to arrive earlier at the Mars cluster than the cargo ships. Once the Mars cluster is reached, the depots located at LDO and LMO are used for refueling. The staggered arrivals of tankers and cargo ships effectively reduce the required depot size in the Mars cluster because ISRU facilities can refill the on-orbit depots in between arrivals.

Table 2 shows all 5 transfer windows that are selected in the first scenario of the baseline case. Notably, windows VI and X are used most extensively because they are among the cheapest interplanetary transfer option with respect to the Δv requirements. About 75% of all cargo is transported through them. The usage of window V, which is a faster but also a more expensive trajectory than window VI, is surprising at first glance because its shorter TOF provides no advantage. Both transfer windows, V and VI, reach the martian cluster before the first demands ever occur (first demand in $t = 9$), rendering speed irrelevant in this particular case. However, transfer window V is being used despite its higher Δv requirements. The reason is that this strategy spreads out the arrivals of vehicles at the Mars cluster ($t_V^{\text{arrival}} = 7$ and $t_{VI}^{\text{arrival}} = 8$). This reduces the total amount of propellant required for refueling the arriving spacecraft in each period at depots. Consequently, this allows for smaller ISRU facilities and depots to be established in the Mars cluster.

While transfer windows V and VI are associated with first-stage decisions, windows IX and X are situated on the second stage. Thus, the nonanticipatory constraints do not force the

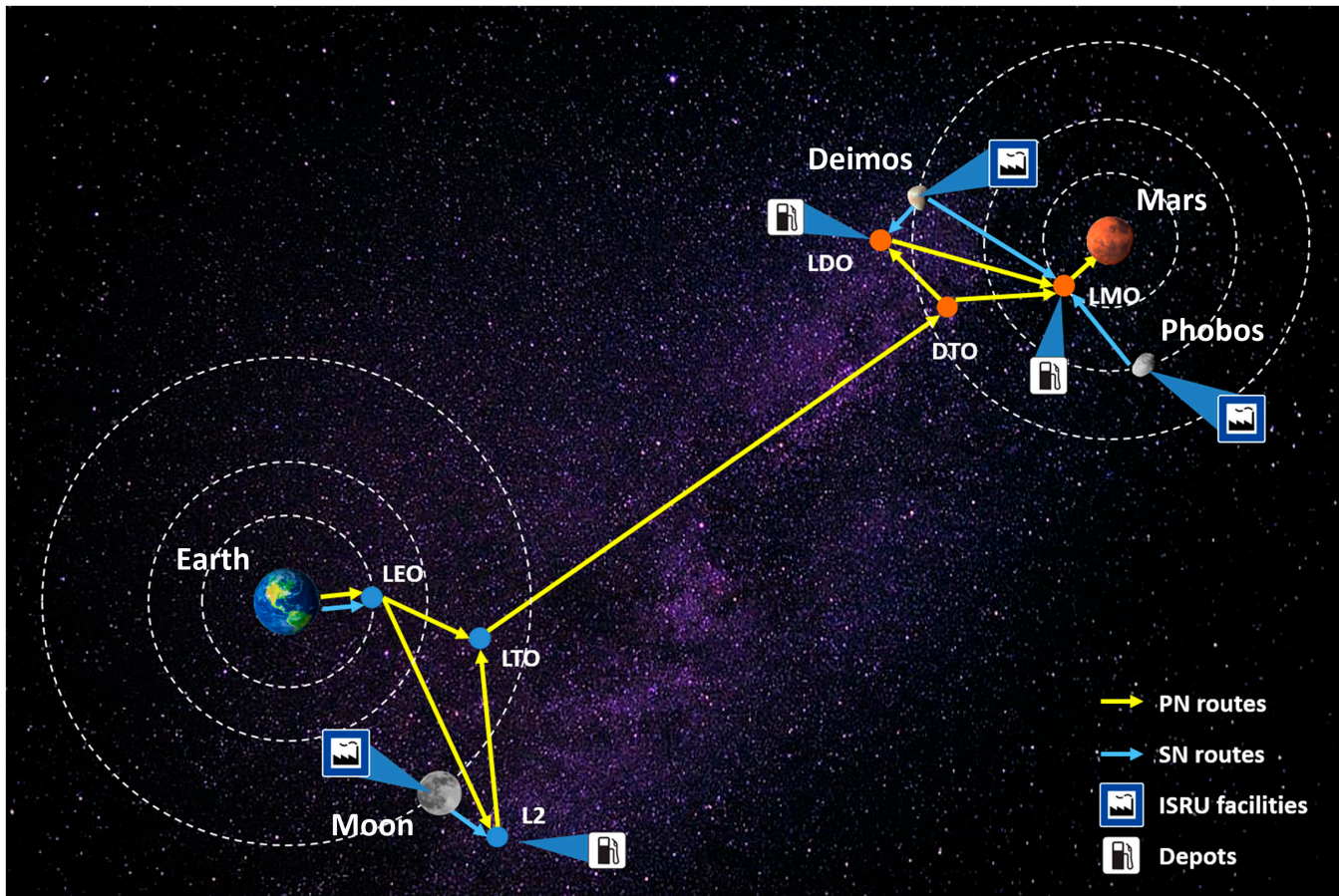


Fig. 4. Complete infrastructure setup and routings through supply chain (SC) network.

Table 2. Transfer windows used by R spacecraft to transport a total of Q tons of cargo in S1 of the baseline case.

Window	$t_{\text{departure}}$	t_{arrival}	R	$Q[t]$
V	4	7	1	29.24
VI	4	8	3	300.00
IX	14	15	1	93.40
X	14	16	3	300.00
XI	19	23	1	78.68

corresponding decision to apply in all scenarios. Similar to the first 2 transfer windows, the trajectories of windows IX and X exhibit the same $t_{\text{departure}}$ but different t_{arrival} , leading to differing TOFs. Spreading out the arrival times provides the same benefits in terms of propellant supply per period that were discussed earlier. The cheaper but slower transfer window X is used to supply a large and consistent amount of payload to the Martian base, whereas the faster and more costly window IX is utilized to transport variable quantities specific to the scenario subsets of the second stage. The last spacecraft in the baseline case is launched during transfer window XI, which has, by far, the highest Δv requirements of all utilized trajectories. However, it is also associated with third-stage decisions and, thus, with

the highest degree of freedom. Hence, the transfer decisions can be made on the scenario level, which allows to respond to specific demand variations.

Because the limited availability of transfer windows prevents a continuous supply of the martian base, inventories are accumulated to bridge the gaps between deliveries. Figure 5 shows the inventories over time including the departure and arrival periods of the utilized transfer windows. It visualizes that the majority of cargo in each stage (or synodic period) is supplied in one large batch using the most propellant efficient trajectories available (i.e., window VI in the first stage, window X in the second stage, and window XI in the third stage). Therefore, inventory levels exhibit a lot of variation over time. The 2 prominent peaks ($t = 8$ and $t = 16$) indicate that it is most efficient to have few but substantial supply runs, which cover almost all demands of the synodic period ahead. The last inventory peak ($t = 23$) is, by far, the smallest, which can be attributed to end of horizon effects.

The total propellant production by each installed ISRU facility and on Earth can be viewed in Fig. 6 (for scenario 1). The depicted bar graph refers to the propellant that is manufactured exclusively for on-orbit refueling. Hence, the propellant manufactured on Earth to fuel spacecraft on the ground before liftoff is omitted in the graphic. Nevertheless, a substantial amount of propellant designated for on-orbit refueling is produced on Earth. As shown in Fig. 4, this propellant is supplied to LEO even though no depot is allocated there. This is possible because depots are only required if propellant is

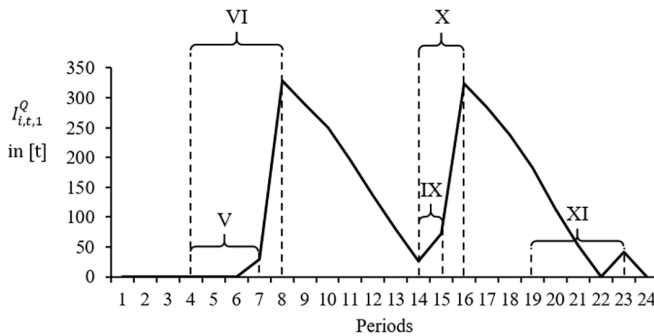


Fig. 5. Inventories on Mars in reference to utilize transfer windows in S1.

stored across time periods. If that is not the case, then spacecraft that are en route to Mars are directly refueled by the vehicles that transport the propellant to their location. It is noteworthy that the Starship, which served as a reference point for many parameters in the baseline case, is being designed to possess this spacecraft-to-spacecraft refueling capability. This strategy is especially useful when Earth is the origin of the propellant supplies because virtually unlimited quantities can be manufactured and supplied in the same time period. Propellant production on other celestial bodies, however, presupposes the installation of costly ISRU facilities, which constraints the production volume. Notably, the capability to store propellant allows to reduce the size of these expensive ISRU plants because the production volume can be spread out over multiple periods. Hence, the allocation of ISRU facilities at the Moon, Phobos, and Deimos is associated with the allocation of on-orbit depots.

Spacecraft parameters

In this section, sensitivity analyses of various spacecraft parameters are conducted. The parameters used in the baseline case are based on spacecraft that are currently in development. However, it can be expected that, by the time ISCs may be attainable, technological capabilities will be far more advanced. This would translate into different spacecraft parameter combinations compared to the baseline case. To explore the effects of the technological progress on the SC network, sensitivity analyses with regard to the spacecraft's payload capacity (cap^Q), its propellant capacity (cap^P), and the specific impulse (I_{sp}) of the used propulsion system are carried out. To this end, the factor ω is applied, which is used to increase one of the respective parameters at a time in 10% increments to a maximum of 200% of the original value (i.e., $\omega = 2.0$). It is assumed that technological progress will only lead to improved parameter settings, and thus, a unidirectional sensitivity analysis only considering parameter increases (and no decreases) is conducted.

Figure 7A shows the TLM for the sensitivity analyses of all 3 parameters in question. The results reveal that the parameter variations impact the OFV very differently. Increasing the cap^Q displays the smallest effects but, nevertheless, leads to a steady decrease of TLM. A similar trajectory results from increasing cap^P . The trend, however, expresses a much steeper decline compared to cap^Q , leading to almost twice the mass reductions with respect to the OFV. Notably, increasing values of I_{sp} have, by far, the greatest impact on the TLM. The corresponding graph follows a clear exponential decline indicating that major gains could be achieved by relatively small improvements in propulsion technology.

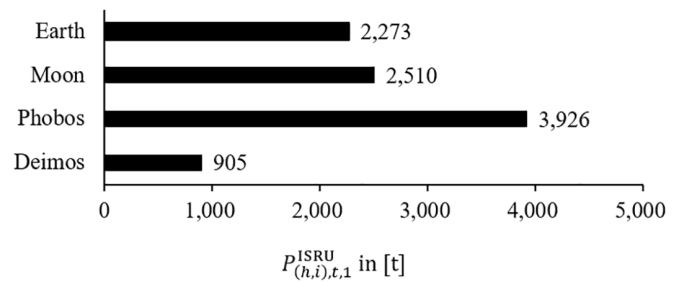


Fig. 6. Total in situ resource utilization (ISRU) propellant production for on-orbit refueling.

In the following, infrastructure decisions are investigated with regards to the changing parameter inputs. Figure 7B shows the total masses of the installed ISRU facilities and depots given different spacecraft payload capacities. At first, depot size is reduced substantially but then remains rather consistent for higher values of ω . The downsizing of the ISRU facilities trails behind the depot size reduction and expresses a more constant decline. However, the masses of both infrastructure elements decline with increasing payload capacities.

A quite different picture emerges when propellant capacities are varied (Fig. 7C). While depot size tends to decline, the installed ISRU facilities appear to slightly increase in mass. This results in diverging trajectories for both infrastructure elements given the increasing values of ω . This development indicates that less propellant is stored at depots over consecutive periods. Instead, the on-orbit propellant supply becomes more immediate as larger ISRU facilities are able to produce and deliver fuel in the same period it is required by spacecraft.

The largest effects on infrastructure decisions result, once again, from changes in the I_{sp} (Fig. 7D). The sizes of both ISRU facilities and depots follow a synchronous exponential decline that very much resembles the trajectory of the corresponding TLM. A higher specific impulse naturally lowers the propellant requirements for all routes, which allows for drastic downsizing of the required infrastructure. Moreover, it infers more advanced propulsion technologies, which may include novel chemical propulsion systems or completely different options such as nuclear thermal or solar electric propulsion [13]. As indicated by these results, the improvement of propulsion systems could be key to actually make ISCs a reality.

ISRU availability

In the baseline case, all surface nodes serve as possible locations for ISRU facilities. This presupposes the technological capabilities and sufficient resource prevalence on each site. In this section, additional cases in which ISRU is not possible at every single surface node are explored and compared to the baseline case. These cases of limited ISRU availability are intended to reflect different stages of technological progress. The first instances assume no ISRU capabilities at all, meaning that only Earth may provide propellant. In the second instance, the Moon provides the only option for ISRU activities. Given the inherent complexity of installing, operating, and maintaining such facilities, it is reasonable to assume that proximity to Earth will be important for the first sizable implementations of this technology. However, the first ISRU capabilities in the Mars cluster are likely to be established on the Red Planet in the vicinity of the base. This facilitates operations and maintenance of such

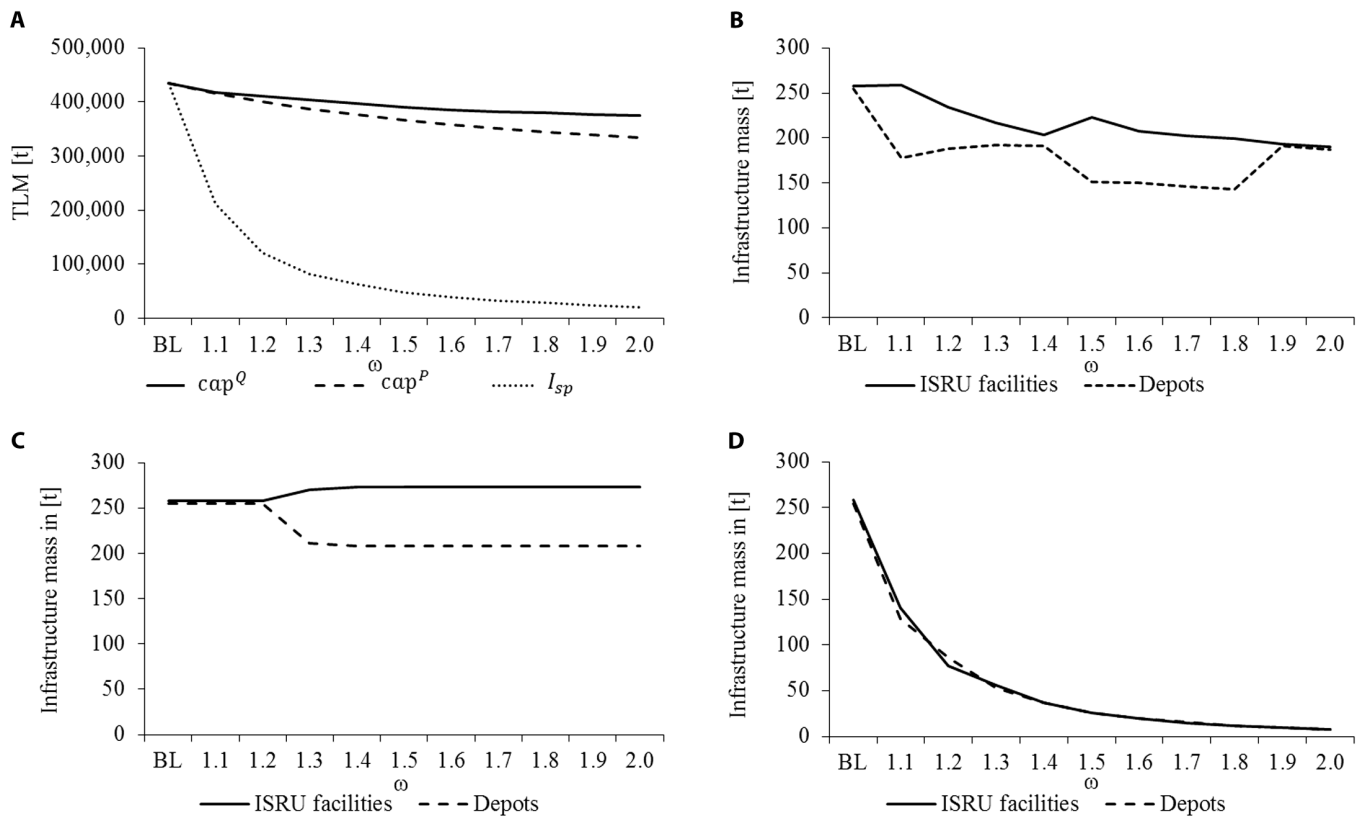


Fig. 7. (A) Total launch mass (TLM) for varying cap^Q , cap^P , and I_{sp} . Mass of ISRU facilities and depots for varying (B) cap^Q , (C) cap^P , and (D) I_{sp} .

facilities while it also assists with sustaining the base itself. Hence, the third case assumes that, in addition to the Moon, ISRU facilities may be set up on Mars. Corresponding technologies are already being developed and experimented with (e.g., Mars Oxygen In Situ Resource Utilization Experiment or MOXIE in short). Utilizing Deimos and/or Phobos for propellant production will likely be the last capability developed, which represents the fourth and final case (=baseline case). Given that Mars' satellites have no atmosphere and almost no gravity, they might be the most difficult locations to produce propellant [4]. Moreover, scientific evidence is not yet conclusive on whether Phobos and/or Deimos actually possess the resources to manufacture the required amounts of propellant [26]. To sum up, the investigated instances presume the following surface nodes to be available for ISRU allocations: (a) none, (b) Moon only, (c) Moon and Mars, and (d) all (i.e., baseline case).

Figure 8 shows the OFV for each of the considered instances. Notably, having no ISRU opportunities (a) results in an infeasible solution, meaning that demands at the Mars base cannot be fully met throughout the planning horizon. This highlights the key role of ISRU technology to establish an ISC. If propellant production is possible at the Moon only (b), then the problem becomes feasible although the TLM is extremely high compared to the remaining instances. As soon as Mars becomes a viable option for ISRU activities (c), the OFV drops by about 30%. Moreover, having all surfaces available for propellant production (d) reduce the TLM also by 30% in reference to the previous Moon and Mars instance. It is noteworthy that our knowledge about resource availability on Moon and Mars is more advanced than for Deimos and Phobos. Scientific evidence

indicates that both Moon and Mars provide the conditions to produce fuel from surface materials (regolith and/or water ice) and/or the atmosphere [27,28]. Our understanding of Deimos' and Phobos' compositions, on the other hand, is much more limited and requires further exploration [26]. However, the results show that increasing availability of ISRU locations substantially improves the efficiency of the ISC.

Furthermore, the availability of ISRU locations affects the allocation and sizing of the corresponding facilities. As indicated in Table 3, increasing location availability leads to a drastic decline in total ISRU facility mass. At the same time, propellant production becomes much more decentralized as a growing number of sites are selected for ISRU activities. Consequently, the average facility mass per location declines. In other words, increasing location availability promotes the allocation of more but smaller ISRU facilities. Notably, major share of the propellant production is carried out in the Mars cluster as soon as the respective locations are available. This is an interesting finding considering the largest part of the journey is already completed at this point. However, the results show that it is optimal for the spacecraft to refuel upon arrival in the Mars cluster because it reduces the amount of propellant that has to be carried along the intercluster trip.

Conclusion

This work sets out to establish SC planning concepts in the realm of optimization-based space logistics. More specifically, the optimal design of an ISC able to support a permanent base on Mars is determined. To that end, a stochastic multistage

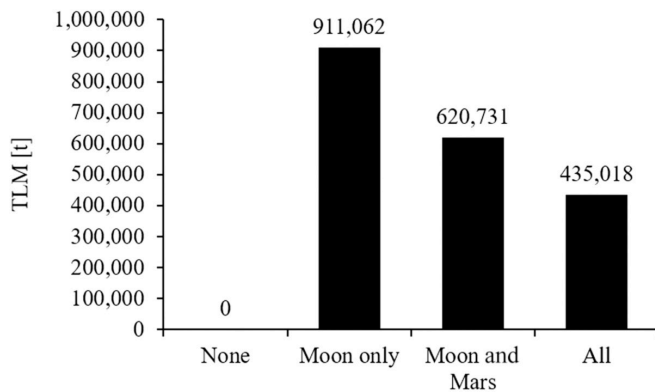


Fig. 8. TLM in all considered instances.

MILP model is developed that integrates production–distribution and in-space infrastructure allocation decisions in the presence of demand uncertainty. The model accounts for orbital dynamics through the concept of a dynamic network with moving nodes. Moreover, a new perspective on fuel management is offered by addressing this subject in the context of ISC planning. This involves key principles like mass-dependent fuel consumption, propellant as a limited resource, and infrastructure allocations for in-space refueling. A layered network structure is considered, consisting of a PN and an SN, which are responsible for cargo and propellant supply, respectively. To validate the model, case studies were constructed based on real data on Earth–Mars transfer windows and flight trajectories obtained through NASA's trajectory browser. Using these cases, extensive numerical analyses were carried out in order to identify the impacts of various stages of technological progress on the optimal ISC design.

The results of the numerical analyses are characterized by frequent spacecraft interactions and the sophisticated coordination of vehicle fleets in conjunction with the propellant supply network. This highlights the necessity of holistic SC planning approaches to support continuous deep space operations. More specifically, the findings show that a division-of-labor approach is often applied in which some spacecraft carry the bulk of the payload while others fetch propellant from on-orbit depots to supply the entire fleet. Furthermore, spacecraft that launched together from Earth would utilize different transfer windows in order to stagger their arrival times. This effectively allows for smaller-sized ISRU facilities and depots in the Mars cluster because less propellant needs to be provided per period. The results also reveal that advancements in the field of propulsion technology would greatly impact the optimal network design while progress with regards to other spacecraft properties has lesser effects. Moreover, the availability of ISRU locations strongly influences the optimal solution, indicating that the technological capability to manufacture propellant on other celestial bodies is key to an efficient ISC. Overall, the proposed model provides valuable insights and expands our understanding of SC planning by a space logistics perspective.

This work is intended to lay the foundation for further research in the vast and mostly uncharted field of ISC planning. While the proposed model considers demand uncertainty, other sources of uncertainty might have network shaping effects and are, thus, worth exploring as well. This might include operational risks such as production yields of ISRU facilities or disruption risks

Table 3. ISRU facility allocation and sizing in [t] for varying location availability.

	Available ISRU locations			
	None	Moon only	Moon and Mars	All
Moon	-	1,143	131	112
Mars	-	0	130	0
Deimos	-	0	0	22
Phobos	-	0	0	124
Total	-	1,143	261	258

like machine failures or even accidents. Given the vast distances and corresponding TOF, the responsiveness of ISCs with regards to any kind of risk is generally very limited. Hence, it is paramount to account for uncertainties on a strategic planning level and further study their effects on the optimal network design.

Moreover, the developed model focuses on material deliveries to Mars and omits the transport of astronauts to the Red Planet. Notably, manned space flights have a number of different requirements. For instance, they demand transfer windows with short TOFs, they have different capacity restrictions, and on-board resource consumption also needs to be considered. Given these characteristics, future works might include corresponding constraints to study the impact of manned missions on the ISC network design. Furthermore, the proposed case studies are very much based on projections of parameters and properties because many of the respective technologies are not yet available. For instance, the size and mass of on-orbit depots and ISRU facilities can only be estimated from the scientific literature at this point (e.g., [6,29]). As the technological progress continues, however, more accurate information and knowledge will become available, which can be utilized to advance corresponding case studies. This may also include novel propulsion technologies such as solar electric propulsion or nuclear electric propulsion. To date, these propulsion systems are characterized by high fuel efficiency but low thrust, which results in very long transportation times [30]. These technologies might be good options for transporting supplies, but they are likely not viable for human space flight because time is much more critical here. Therefore, it is worth exploring how these technology options in combination with the transported cargo impact the optimal ISC design. Another important property that could be studied in future works is the maintenance of the in-space infrastructure. Structures like on-orbit depots and ISRU facilities may require regular services in order to ensure their functioning. This might promote infrastructure allocations in Earth's relative vicinity because maintenance efforts are much easier to conduct there.

In any case, research on these issues can be expected to be of growing importance. Rapid technological advancement and innovation have opened up evermore commercial opportunities in space, which has drawn increasing public attention and pushed the topic on the political agenda. The powerful alignment of interests in the fields of science, economy, and politics is a historically rare phenomenon with the potential to accelerate the progress in the space sector even further. Therefore, it is

vital to lay the theoretical foundation and explore new logistics paradigms that enable sustainable deep space exploration.

Acknowledgment

Competing interests: The author declares that there are no conflicts of interest regarding the publication of this article.

Data Availability

The data used to support the findings of this study are available upon reasonable request.

Supplementary Materials

Sections S1 to S4

Tables S1 to S5

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A Stochastic Modeling Approach for Interplanetary Supply Chain Planning

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The unique and complex challenges of sustainable interplanetary travel necessitate a novel space logistics paradigm, which is hereafter referred to as the interplanetary supply chain. To lay the groundwork for this anticipated paradigm shift, this work aims to introduce principles and methodologies of supply chain (SC) planning to the field of space logistics. To this end, a stochastic multistage mixed integer linear programming model is proposed that explores the vision of a permanent human presence on Mars from a logistics standpoint. The model optimizes material flows, propellant production, and in-space infrastructure allocations to identify the optimal SC network design. Notably, demand uncertainty at the Mars base is considered, which makes it the first model to investigate the effects of stochasticity in space logistics planning. Several case studies are investigated, which incorporate data on Earth–Mars transfer windows and flight trajectories obtained through the National Aeronautics and Space Administration's trajectory browser. The results reveal numerous interdependencies and, thus, highlight the need for holistic SC modeling approaches that go beyond the isolated mission planning of the extant literature. For instance, the findings show frequent interactions among spacecraft as they regularly engage in division of labor regarding cargo and propellant transportation. Moreover, complex propellant supply networks that involve strategic in-space infrastructure allocation and propellant production on celestial bodies are established. Overall, the developed model provides valuable insights into the planning of a sustainable interplanetary travel, offers a new SC perspective, and lays the foundation for future research in this area.

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